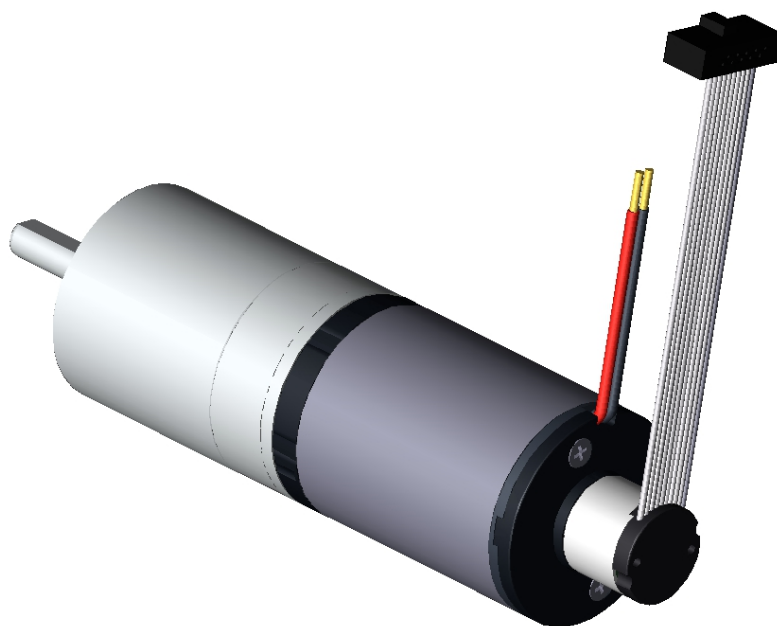


Configured drive

Motor - DC-max22S GB KL 12V
Gearhead - GPX22 C 35:1
Sensor - ENX10 EASY 1024IMP

Part number: B82D23AD645E Revision number 3

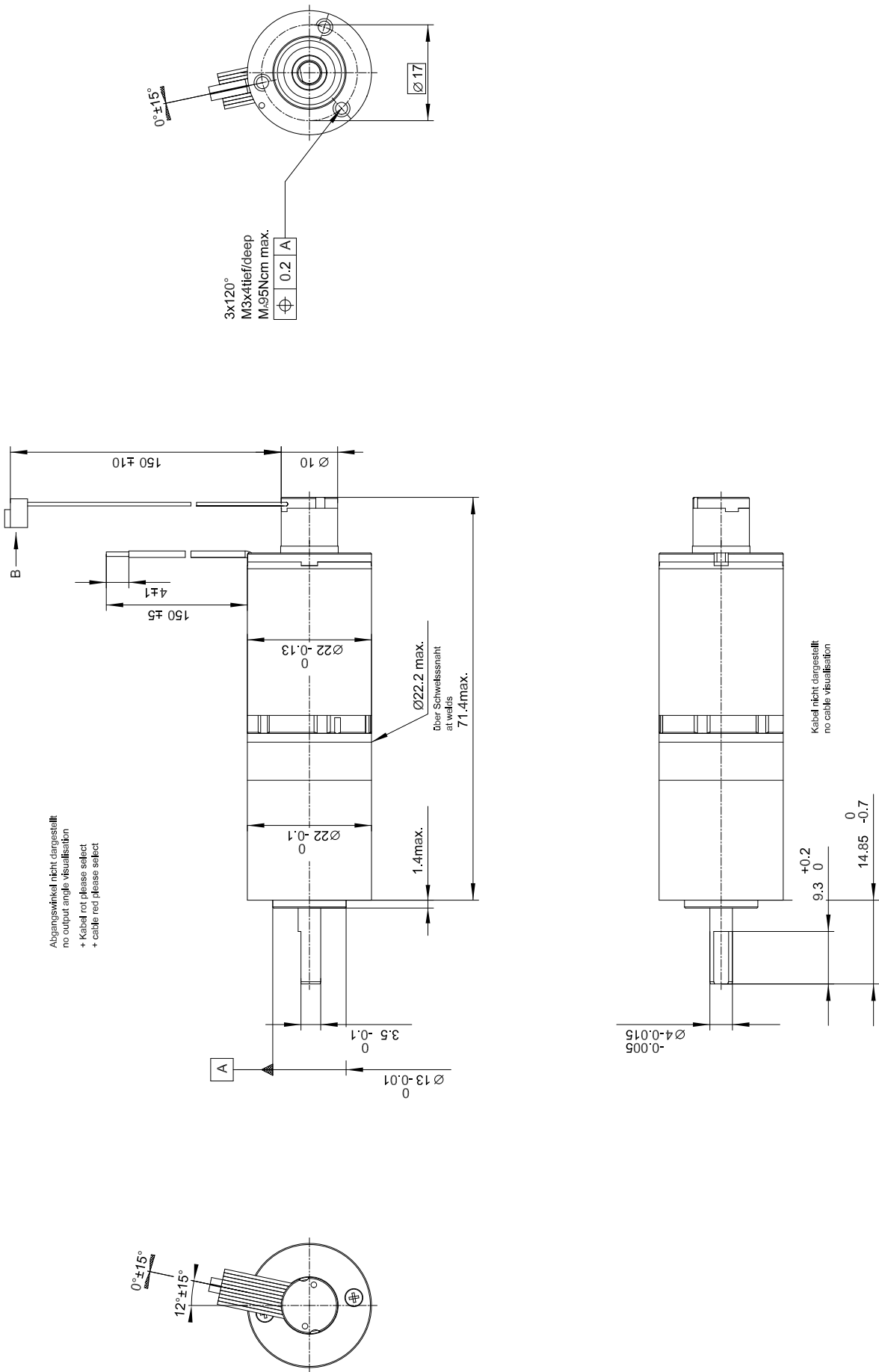
General Terms and Conditions: https://www.maxongroup.ch/maxon/view/content/terms_and_conditions_page

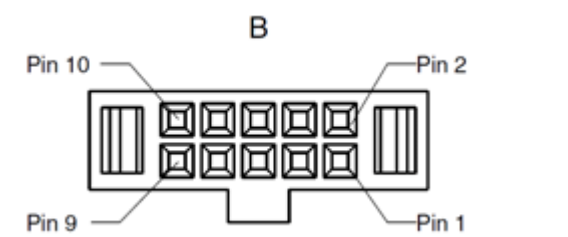


To open the integrated CAD file, please save this document and open it in Acrobat Reader. The STEP file is available after a double-click on the pin icon.

B82D23AD645E.stp (STP AP 214)

Open configuration: <https://www.maxongroup.com/maxon/view/configurator/?ConfigID=B82D23AD645E>





Connector type, encoder	
1.27mm 10-pol	
Pin 1	Do not connect
Pin 2	VCC
Pin 3	GND
Pin 4	Do not connect
Pin 5	Channel A \
Pin 6	Channel A
Pin 7	Channel B \
Pin 8	Channel B
Pin 9	Channel I \
Pin 10	Channel I

Summary of your selected configuration

Total weight of the drive: 110 g

DC-max22S GB KL 12V

Product detail	
Commutation	Graphite brushes
Nominal voltage	12
Motor bearings	Ball bearings
Elec. connection, motor	
Electrical connection, motor	Cable
Cable length	150

GPX22 C 35:1

Product detail	
Gearhead type	CA
Reduction	35
Number of stages	2

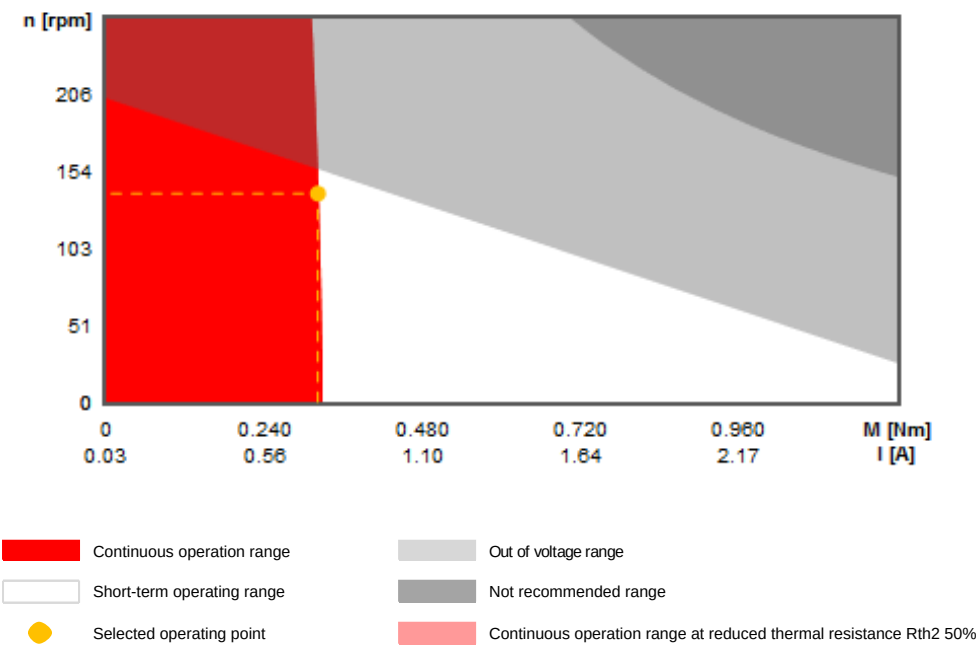
ENX10 EASY 1024IMP

Product detail	
Counts per turn	1024
Connector type, encoder	1.27mm 10-pol

Legend for part designation

EC-Motor:		DC-Motor:		IDX Drive/Motor:	
A	Hall sensor version	CLL	Capacitor long life (spark suppression)	AB	Integrated brake
B	Sensorless version	DC-max	DC motors (brushed)	CO	CANOpen interface
BL	Brushless	DCX	DC motors (brushed)	ENC	Encoder
C	Integrated electronics	EB	Precious metal brushes	ET	EtherCAT interface
ECX	EC-motor (brushless)	GB	Graphite brushes	IDX	Drive System
FL	FLAT (type series name)	KL	Ball bearings	IO	I/O interface
HP	High Power	S/M/L	Short / Medium / Long	P	Positioning-control
HTQ	High Torque	SL	Sintered sleeve bearings	S	Speed-control
HTQF	High Torque, fan-ventilated			S/M/L	Short / Medium / Long
KL	Ball bearings				
M/L/XL	Medium / Long / Extra long				
PR	PRIME (type series name)				
SP	SPEED (type series name)				
STD	Standard				
STE	Sterilizable				
STEC	Sterilizable, ceramic bearing				
TQ	TORQUE (type series name)				
Gearhead		Encoder/Sensor:			
A	Standard version	ABS	Absolute		
C	Ceramic version	BiSSC	Bidirectional serial synchronous (protocol)		
GPX	Planetary gearhead	COMM	Commutation signals		
HP	High Power	EASY	(type series name)		
LN	Reduced noise level	EMT	(type series name)		
LZ	Reduced backlash	ENX	Encoder		
SP	SPEED (type series name)	FFC	Flexible flat cable		
ST	Number of stages	GAMA	(type series name)		
STE	Sterilizable	IMP	Pulses		
STES	Sterilizable, with sealing	INT	integrated		
UP	Ultra Performance	MAG	(type series name)		
		QUAD	(type series name)		
		RIO	(type series name)		
		SSI	Synchronous serial Interface (protocol)		
		XT	Extended version		

Drive disposition



Combination details

Your entries

Available voltage	12 V
Speed	140 min ⁻¹
Torque	0.321 Nm

Values of the drive at available voltage

Available voltage	12 V
Max. speed at given load	155 min ⁻¹
Max. continuous torque	0.35 Nm
Max. continuous current	0.82 A

Required electrical data for your operating point

Speed	140 min ⁻¹
Torque	0.321 Nm
Required voltage	11.01 V
Required current	0.74 A

DC-max22S GB KL 12V

Product specification

Values at nominal voltage

Nominal voltage	12 V
No load speed	7140 min ⁻¹
No load current	28.3mA
Nominal speed	5260 min ⁻¹
Nominal torque (max. continuous torque)	12.5 mNm
Nominal current (max. continuous current)	0.819 A
Stall torque	49.4 mNm
Stall current	3.14 A
Max. efficiency	80.6 %

Characteristics

Max. output power continuous	10.4 W
Terminal resistance	3.83 Ω
Terminal inductance	0.274 mH
Torque constant	15.8 mNm A ⁻¹
Speed constant	606 min ⁻¹ V ⁻¹
Speed/torque gradient	147 min ⁻¹ mNm ⁻¹
Mechanical time constant	8.57ms
Rotor inertia	5.57 gcm ²

Thermal data

Thermal resistance housing-ambient	20 KW ⁻¹
Thermal resistance winding-housing	6 KW ⁻¹
Thermal time constant of the winding	17.2 s
Thermal time constant of the motor	538 s
Ambient temperature	-30...85 °C
Max. winding temperature	125 °C

Mechanical data

Max. permissible speed	9000 min ⁻¹
Axial play	0.05...0.15 mm
Preload	0 N
Radial backlash	0.025 mm
Max. axial load (dynamic)	3.3 N
Max. force for press fits (static)	45 N
Static, supported shaft	420 N
Max. radial load 5 mm from flange	12.3 N
Measurement from the flange	5 mm

Further specifications

Number of pole pairs	1
Number of commutator segments	9

Motor weight	46.9 g
Motor length	33.2 mm
Typical noise level	55 dBA (6000 min ⁻¹)

More information on the motor: <https://www.maxongroup.com/maxon/view/product/DC-MAX22S01GBKL598?download=show>

GPX22 C 35:1

Product specification

Gearhead data

Reduction	35:1
Absolute reduction	4554/130
Number of stages	2
Max. continuous torque	0.90 Nm
Max. intermittent torque	1.2 Nm
Direction of rotation, drive to output	=
Max. efficiency	81 %
Average backlash no-load	1.1 °
Mass inertia	0.501 gcm ²
Max. transmittable power (continuous)	15 W
Max. short-time transferable output	19 W

Technical data

Output shaft bearing	Ball bearings
Max. radial play, 5 mm from flange	max. 0.2 mm
Axial play	0...0.2 mm
Max. permissible radial load, 5 mm from flange	100 N
Max. permissible axial load	40 N
Max. permissible force for press fits	100 N
Max. continuous input speed	10000 min ⁻¹
Max. intermittent input speed	12500 min ⁻¹
Recommended temperature range	-40..100 °C

More information on the gearhead: <https://www.maxongroup.com/maxon/view/product/GPX22CAKLSL35D0CPLW?download=show>

ENX10 EASY 1024IMP

Product specification		
Sensor data		
	Counts per turn	1024
	Number of channels	3
	Line Driver	RS422
	Max. electrical speed	90000 min ¹
	Max. mechanical speed	30000 min ¹
Technical data		
	Supply voltage Vcc	5 V ±10 %
	Output signal	INC
	Output signal driver	Differential / EIA RS 422
	Output current per channel	-20...20 mA
	State length	20...160 °el
	Signal rise time/Signal fall time	20/20 ns
	Min. state duration	125 ns
	Direction of rotation	A before B CW
	Index position	A low and B low
	Index synchronously to AB	Yes
	Index pulse width	90 °el
	Typical current draw at standstill	22 mA
	Max. moment of inertia of code wheel	0.05 gcm ²
	Operating temperature range	-40...100 C°
	Number of autoclave cycles	0

More information on the encoder: <https://www.maxongroup.com/maxon/view/product/ENX10EASY06?download=show>